

Barycentric Projection: Application

Optimal Transport for Domain Adaptation

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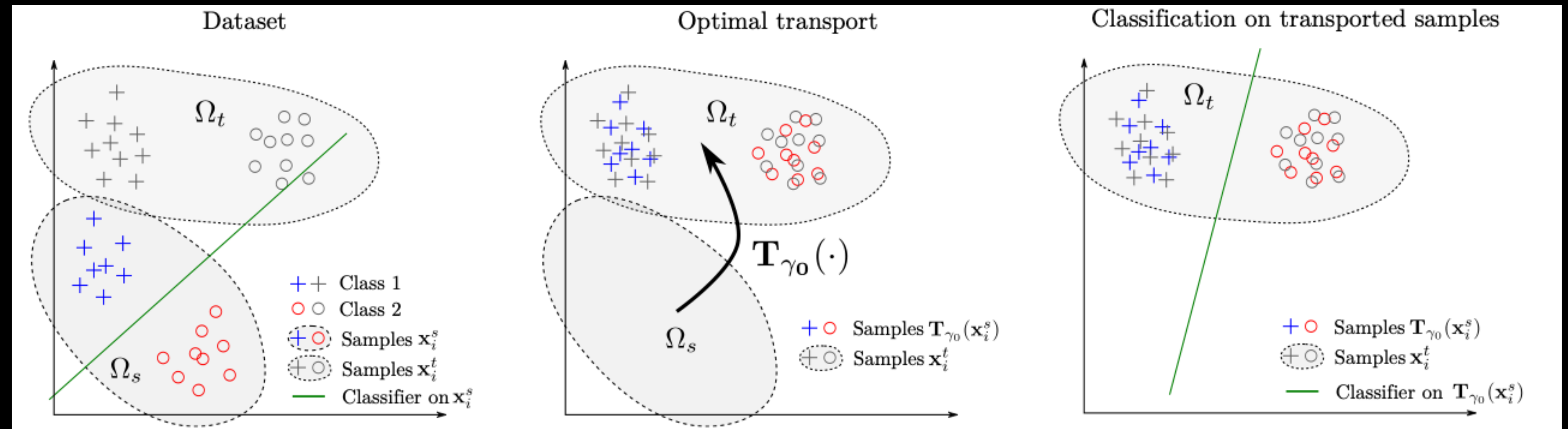
Abstract—Domain adaptation is one of the most challenging tasks of modern data analytics. If the adaptation is done correctly, models built on a specific data representation become more robust when confronted to data depicting the same classes, but

This transformation can be conveniently expressed with respect to the target samples as the following **barycentric mapping**:

$$\widehat{\mathbf{x}}_i^s = \operatorname{argmin}_{\mathbf{x} \in \mathbb{R}^d} \sum_j \gamma_0(i, j) c(\mathbf{x}, \mathbf{x}_j^t). \quad (13)$$

where $\mathbf{x}_i^s$ is a given source sample and $\widehat{\mathbf{x}}_i^s$ is its corresponding image. When the cost function is the squared ℓ_2 distance, this barycenter corresponds to a weighted average and the sample is mapped into the convex hull of the target samples. For all source samples, this barycentric mapping can therefore be expressed as:

$$\hat{\mathbf{X}}_s = \mathbf{T}_{\gamma_0}(\mathbf{X}_s) = \operatorname{diag}(\gamma_0 \mathbf{1}_{n_t})^{-1} \gamma_0 \mathbf{X}_t. \quad (14)$$



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